



Proteomic analysis of exosome-enriched fractions derived from cerebrospinal fluid of amyotrophic lateral sclerosis patients

Noriko Hayashi^a, Hiroshi Doi^{a,*}, Yoichi Kurata^b, Hiroyuki Kagawa^b, Yoshitoshi Atobe^c, Kengo Funakoshi^c, Mikiko Tada^a, Atsuko Katsumoto^a, Kenichi Tanaka^a, Misako Kunii^a, Haruko Nakamura^a, Keita Takahashi^a, Hideyuki Takeuchi^a, Shigeru Koyano^a, Yayoi Kimura^b, Hisashi Hirano^b, Fumiaki Tanaka^{a,*}

^a Department of Neurology and Stroke Medicine, Yokohama, Japan

^b Advanced Medical Research Center, Yokohama, Japan

^c Department of Neuroanatomy, Yokohama City University Graduate School of Medicine, Yokohama, Japan

ARTICLE INFO

Article history:

Received 27 August 2019

Received in revised form 15 October 2019

Accepted 21 October 2019

Available online 24 October 2019

Keywords:

Exosome

Amyotrophic lateral sclerosis

Cerebrospinal fluid

Mass spectrometry

ABSTRACT

Exosomes contain many proteins associated with neurodegenerative diseases. To identify new candidate biomarkers and proteins associated with amyotrophic lateral sclerosis (ALS), we performed liquid chromatography-tandem mass spectrometry proteomic analysis of exosome-enriched fractions isolated from cerebrospinal fluid (CSF) of sporadic ALS patients using gel filtration chromatography. Proteomic data revealed that three proteins were increased and 11 proteins were decreased in ALS patients. The protein with the greatest increase in exosome-enriched fractions of CSF derived from ALS was novel INHAT repressor (NIR), which is closely associated with nucleolar function. By immunohistochemical analysis, we found that NIR was reduced in the nucleus of motor neurons in ALS patients. Our results demonstrate the potential utility of our methodology for proteomic analysis of CSF exosomes and suggest that nucleolar stress might play a role in sporadic ALS pathogenesis through the dysfunction of NIR.

© 2019 Elsevier B.V. and Japan Neuroscience Society. All rights reserved.

1. Introduction

Amyotrophic lateral sclerosis (ALS) is an adult-onset neurodegenerative disease characterised by progressive muscle weakness and atrophy and death within a few years (Taylor et al., 2016). At present, the diagnosis of ALS is made based on clinical findings in accordance with the Revised El Escorial diagnostic criteria (rEEC) (Brooks et al., 2000). However, no effective treatment is available to halt the progression of this disease. The essential pathological features of ALS are progressive loss of motor neurons in the brain and spinal cord and the formation of neuronal cytoplasmic inclusions (NCIs) in motor neurons. NCIs are composed of misfolded proteins including transactive response DNA-binding protein 43 (TDP-43) as a major component in most cases.

Exosomes are extracellular microvesicles (30–150 nm in diameter) secreted by multiple cell types (Patel et al., 2019). They are involved in the clearance or cell-to-cell spreading of molecules such as proteins, lipids, and nucleic acids. An increasing number of studies suggest that exosomes play various roles in the pathogenesis of neurodegenerative diseases. For example, cerebrospinal fluid (CSF) exosomes from patients with Parkinson's disease and dementia with Lewy bodies contain pathogenic species of α -synuclein and induce the oligomerisation of α -synuclein in vitro (Grey et al., 2015; Stuenkel et al., 2016). In Alzheimer's disease (AD), exosomes from neurons or microglia contribute to the clearance of amyloid beta (Yuyama et al., 2012, 2015), and hyperphosphorylated misfolded tau is contained in the CSF exosomes of AD patients (Saman et al., 2012). TDP-43 has also been reported to be transmitted from cell to cell via exosomes in ALS (Ding et al., 2015; Iguchi et al., 2016).

Considering that exosomes derived from the CSF of patients with neurodegenerative diseases are disease-specific, they may contain a subset of proteins reflecting each disease condition, the roles of which are still unknown. In contrast to total CSF containing various blood-derived proteins or CSF nonexosomal fractions, components of CSF exosomes show high enrichment of brain-derived proteins (Chiasserini et al., 2014). Therefore, the analysis of CSF exosomes

* Corresponding authors at: Department of Neurology and Stroke Medicine, Yokohama City University Graduate School of Medicine, 3-9 Fukuura, Kanazawa-ku, Yokohama 236-0004, Japan.

E-mail addresses: hdoi@yokohama-cu.ac.jp (H. Doi), ftanaka@yokohama-cu.ac.jp (F. Tanaka).

might be particularly advantageous for the identification of proteins that reflect the underlying neuropathology.

The present study aimed to identify new candidate biomarkers and/or disease-associated proteins of ALS through the analysis of CSF exosomes. We performed liquid chromatography-tandem mass spectrometry (LC-MS/MS) proteomic analysis of exosome-enriched fractions isolated from the CSF of sporadic ALS patients to reveal the protein profile of CSF exosomes. Our results revealed novel INHAT repressor (NIR) as a protein that was increased in exosome-enriched fractions from the CSF and pathologically confirmed that NIR expression was reduced in motor neurons of sporadic ALS patients, suggesting NIR as a candidate biomarker protein for ALS.

2. Materials and methods

2.1. Human CSF samples

For LC-MS/MS analysis of exosomes, CSF samples were obtained by lumbar puncture from three sporadic ALS patients and three idiopathic normal-pressure hydrocephalus (iNPH) patients as gender- and age-matched disease controls (Table 1). All sporadic ALS patients were diagnosed in accordance with the rEEC (Brooks et al., 2000), and they had neither a family history of ALS nor C9orf72 hexanucleotide repeat expansion confirmed by repeat-primed PCR analysis. For western blot analysis and electron microscopy, a CSF sample was obtained from a 75-year-old female iNPH patient. Following collection, all CSF samples were centrifuged at 500g for 10 min at 4°C, and the supernatants were stored at –80°C until exosome isolation. This study was approved by the Institutional Review Board of Yokohama City University School of Medicine (B140904003), and written informed consent was obtained from all of the patients. We adopted several methods to process CSF samples for each assay as described in the following sections, and the summarised scheme for each process is shown in Fig. 1A.

2.2. Exosome purification from CSF

A total of 3.5 ml of CSF was supplemented with 1% phosphatase inhibitor cocktail 2, 1% phosphatase inhibitor cocktail 3 (Sigma-Aldrich, St. Louis, MO, USA), and Complete Mini Protease Inhibitor 1 tablet/10 ml (Roche Diagnostics, Mannheim, Germany). Samples were centrifuged at 10,000g and 4°C for 30 min, and the supernatants were filtrated with an Ultrafree-CL device, GV0.22 µm (EMD Millipore, Billerica, MA, USA). Filtrated samples were concentrated using Amicon Ultra 100 K columns (EMD Millipore) to a final volume of 100 µl. Exosomes were isolated from the concentrated samples using a Size Exclusion Chromatography on Drip Column (EVSecond) device (GL Science, Tokyo, Japan) (Fig. 1A).

2.3. Exosome sandwich ELISA

The exosome fractions were defined by sandwich ELISA to detect the exosome marker protein CD9 (Fig. 1A, Supplementary Fig. 1A) (Kawakami et al., 2017). A NUNC MaxiSorp plate (Thermo Fisher Scientific, Bremen, Germany) was coated with anti-CD9 antibody in PBS at 4°C overnight. A total of 150 µl of 5% BSA/PBS was added and incubated at room temperature for 30 min with shaking. After washing three times with PBS, 50 µl of PBS and 50 µl of each fraction isolated by the EVSecond device were added and incubated at room temperature for 3 h with shaking. After washing, biotinylated anti-CD9 antibody in 1% BSA/PBS was added in a final volume of 100 µl and incubated at room temperature for 1 h with shaking. After washing, 1:2000 diluted streptavidin-HRP (Vector Laboratories, Burlingame, CA, USA) in 1% BSA/PBS was added to a final volume of 100 µl and incubated at room temperature for 1 h

with shaking. After washing, One-Step Ultra TMB-ELISA Substrate Solution (Thermo Fisher Scientific) was added, and fluorescence intensity was measured at 450 nm using an Infinite 200 fluorescence plate reader (Tecan, Männedorf, Switzerland). The protein concentration was measured using a Qubit fluorometer (Thermo Fisher Scientific).

2.4. Western blot analysis

A total of 4 ml of CSF was centrifuged at 10,000g and 4°C for 30 min, filtrated to 0.22 µm, and concentrated using Amicon Ultra 100 K columns using the same protocol as described above. Half of the concentrated CSF was ultracentrifuged at 100,000g for 70 min at 4°C, and the pellets were washed with PBS, followed by ultracentrifugation at 100,000g for 60 min at 4°C as previously described (Thery et al., 2006). Finally, the pellets were resuspended in SDS sample buffer. The remaining half of the concentrated CSF was subjected to EVSecond isolation by the same protocol as described above, and the obtained fractions were divided into four groups: pre-exosome, pre-exosome peak, exosome peak, and post-exosome peak fractions. Each fraction was then ultracentrifuged at 100,000g for 60 min at 4°C, and the pellets were resuspended in SDS sample buffer (Fig. 1A). Next, 15 µl of CSF and each fraction of the samples were subjected to SDS-PAGE followed by western blot analysis with anti-CD9 and anti-CD81 antibodies.

2.5. Sample preparation for LC-MS/MS

The exosome-enriched fraction samples isolated from CSF were desalted using the 2D clean-up kit (GE Healthcare, Little Chalfont, UK). The obtained pellets were dissolved in lysis buffer (50 mM Tris-HCl pH 7.5, 8 M urea, 2 M thiourea, 4% sodium deoxycholate, 1% phosphatase inhibitor cocktail 2, 1% phosphatase inhibitor cocktail 3, and Complete Mini Protease Inhibitor 1 tablet/10 ml), reduced with 10 mM dithiothreitol for 30 min at 60°C, and alkylated with 10 mM iodoacetamide for 15 min at room temperature. Samples were diluted four times with 50 mM ammonium bicarbonate and digested with trypsin (Promega, Madison, WI, USA) at an enzyme:substrate ratio of 1:10 for 16 h at 37°C. To remove sodium deoxycholate in the peptide solution, digested samples were acidified with 2.5% trifluoroacetic acid (TFA), mixed with the same volume of ethyl acetate, and centrifuged at 15,000g for 1 min at 4°C, and the aqueous phase was collected. Then, the same procedure was repeated. Peptides were desalted using an SPE C-tip, in accordance with the manufacturer's protocol (Nikkyo Technos, Tokyo, Japan). To deglycosylate the peptides, desalted samples were diluted in 50 mM phosphate buffer with 10 mM EDTA and incubated with N Glycosidase F (Roche Diagnostics) for 20 h at 37°C. Deglycosylated peptide samples were then desalted using an SPE C-Tip SDB/CB and were eluted by gradient elution with 70%, 80%, 90%, and 100% acetonitrile/0.1% TFA. Finally, the samples were dried by vacuum centrifugation and dissolved with 0.1% TFA.

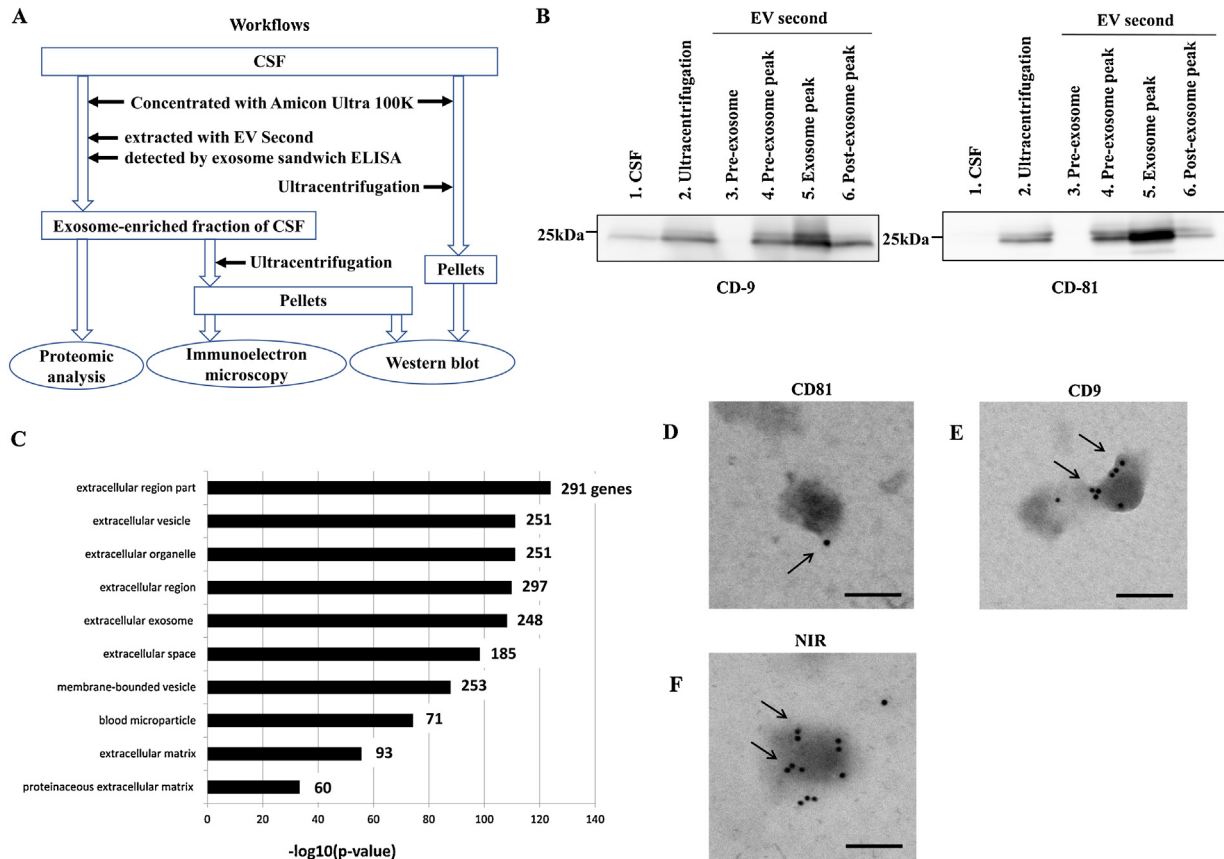
2.6. Proteomic analysis

A total of 5 µg of each digested peptide sample was dissolved in loading buffer (0.1% FA) and analysed using a Dionex UltiMate™ 3000 Nano system coupled via a Nanospray Flex ion source (Thermo Fisher Scientific) to a linear ion trap-orbitrap hybrid mass spectrometer (Orbitrap Elite; Thermo Fisher Scientific). Peptides were trapped on a Nanotrap column (Acclaim PepMap™100 C18 column, 100 µm i.d., 20 mm length, 5 µm particle size, and 100 Å pore size; Thermo Fisher Scientific) and separated on a C18 capillary LC column (Nano HPLC capillary column, 75 µm i.d., 125 mm length, 3 µm particle size; Nikkyo Technos) with a linear 180 min gradient of 2%–33% buffer B (0.1% FA, 95% ACN) at a flow rate of 350 nl/min.

Table 1
CSF samples for proteomic analysis.

	Sporadic ALS group			Disease control group		
Case	ALS1	ALS2	ALS3	iNPH1	iNPH2	iNPH3
Age/sex	79/M	71/M	73/M	74/M	69/M	75/M
rEEC	Probable	Definite	Definite	NA	NA	NA
Disease duration at lumbar puncture, month	6	36	33	NA	NA	NA
ALS-FRS score	43	41	42	NA	NA	NA

F, female; M, male; ALS, amyotrophic lateral sclerosis; iNPH, idiopathic normal-pressure hydrocephalus; ALS-FRS, ALS functional rating scale; NA, not applicable.

**Fig. 1.** Exosome marker proteins and NIR in CSF exosome-enriched fractions isolated by the EVSecond device.

(A) Schema of the isolation method of CSF for each analysis. (B) Western blot analysis of CSF exosomes isolated by ultracentrifugation or the EVSecond device. A total of 15 μ l of CSF was loaded in lane 1. CSF exosomes were isolated by ultracentrifugation (lane 2) or by the EVSecond device from 2 ml of CSF. Exosome fractions isolated by the EVSecond device were divided into four groups: pre-exosome (lane 3), pre-exosome peak (lane 4), exosome peak (lane 5), and post-exosome peak fractions (lane 6).

(C) Gene Ontology analysis of proteins identified in exosome-enriched fractions from CSF. The top 10 Gene Ontology terms in cellular components detected by DAVID (v6.8) software for the 334 protein-coding genes of proteins identified in this study are listed.

(D–F) Immunoelectron microscopic images of vesicles isolated from CSF. Scale bar: 100 nm. These vesicles were immunopositive for CD81 (D), CD9 (E), and NIR (F).

MS spectra were acquired in data-dependent acquisition (top 15) in positive mode in the m/z range of 350–1500. The full-scan MS spectra were set to $5e^5$ with a maximum injection time of 500 ms and resolution of 120,000. Fragmentation of precursor ions was performed by collision-induced dissociation with a collision energy of 35 eV. MS/MS scans were recorded for a maximum injection time of 300 ms to a maximal AGC value of $1e^4$, scan range of 350–1500 m/z with an isolation window of 2.0 m/z , and dynamic exclusion time set to 60 s.

Label-free relative peptide quantitation was performed using the Progenesis Q1 for proteomics (Nonlinear Dynamics, Durham, NC, USA) with the following parameters: charges of +2, +3, +4; and analysis of variance (ANOVA) with a significance threshold of $p < 0.05$. The obtained data were used as search queries against the human protein sequences in the UniProtKB/Swiss-Prot database (ver. 2014.7.fasta) using MASCOT (ver. 2.5.1; Matrix Science, London, UK) with the following parameters: peptide mass tolerance

of ± 5 ppm; fragment mass tolerance of ± 0.5 Da; overall peptide false discovery rate of 1%; and a peptide ion score cut-off of >30 . Gene Ontology analysis was conducted using DAVID Bioinformatics Resources 6.8 (NIAID/NIH).

2.7. Immunoelectron microscopy of CSF exosomes

Purification of the exosome-enriched fractions from CSF using EVSecond isolation was performed by the same protocol as described above. Isolated exosome-enriched fractions from CSF were ultracentrifuged at 100,000 g for 70 min at 4 °C, and the pellets were washed with PBS and ultracentrifuged at 100,000 g for 60 min at 4 °C, as previously described (Thery et al., 2006) (Fig. 1A). The obtained pellets were resuspended with 5 μ l of PBS as the samples for electron microscopy. The samples were fixed with the same volume of 8% paraformaldehyde (PFA) in PBS, incubated for 10 min at room temperature, applied to carbon/formvar-coated electron

microscope grids, and air-dried for 20 min at room temperature. The samples were re-fixed with 1% PFA in PBS for 10 min at room temperature, and blocking was performed with 5% bovine serum albumin (BSA) in PBS for 20 min at room temperature. Immunostaining with primary antibodies, anti-NIR, anti-CD9, and anti-CD81 was performed overnight, followed by the application of 12 nm gold-labelled mouse or rabbit secondary antibody. The samples were stained with 2% uranyl acetate and examined by transmission electron microscopy (H-7500; Hitachi High-Technologies Corporation, Tokyo, Japan).

2.8. Immunohistochemistry

Each spinal cord of sporadic ALS patients and disease control subjects (Supplementary Table 1) was fixed in 20% neutral buffered formalin (Muto Pure Chemicals, Tokyo, Japan) for 7–13 days, embedded in paraffin, and sectioned at a thickness of 6 μ m. For immunohistochemical staining of NIR, cervical, thoracic, and lumbar spinal cord sections of each patient were autoclaved for 10 min at 121 °C and incubated in 1% hydrogen peroxide for 20 min. Sections were immunostained with anti-NIR followed by HRP-conjugated anti-mouse IgG antibody using the avidin-biotin-peroxidase complex method (ABC Elite; Vector Laboratories) and were stained with haematoxylin. For double immunofluorescence staining (Supplementary Fig. 2), anti-NIR and anti-TDP-43 primary antibodies were visualised by fluorescence-labelled secondary antibodies.

To evaluate the frequency of NIR positivity of motor neurons, the number of anterior horn neurons with NIR staining of the nucleus was counted in two sections from the thoracic or lumbar spinal cord, as listed in Supplementary Table 1. Statistical analysis was performed with Student's t-test. Differences were considered significant at $p < 0.05$.

3. Antibodies

The antibodies used in this study are listed in Supplementary Table 2.

4. Results

4.1. Isolation of CSF exosome-enriched fractions

First, we validated the methodology for isolating the CSF exosomes and confirmed that the EVSecond, a size exclusion chromatography open column, was appropriate for proteomic analysis in this context. CSF exosomes isolated by the EVSecond contained exosome marker proteins sufficiently resembling those isolated by conventional ultracentrifugation (Fig. 1B).

To identify the relevant fractions of CSF exosomes isolated using the EVSecond device, we performed sandwich ELISA by employing anti-CD9 antibody as described in the Materials and Methods section (Supplementary Fig. 1A). Since the CD9 absorbance peak was observed in fraction 5, we used fractions 4–6 as the exosome-enriched samples for the LC-MS/MS proteomic study. We also generated an ion intensity map of MS data derived from exosome-enriched fractions isolated from the CSF of ALS and iNPH patients using Progenesis Q1 software (Supplementary Fig. 1B) and confirmed that the alignment quality score was sufficient: 89.1% (ALS 1 compared with iNPH 3), 86.6% (ALS 2 compared with iNPH 3), 91.6% (ALS 3 compared with iNPH 3), 92.5% (iNPH 1 compared with iNPH 3), and 92.5% (iNPH 2 compared with iNPH 3). These data suggested that we had successfully isolated homogeneous CSF exosome-enriched samples from individual patients.

4.2. Proteomic analysis of proteins in exosome-enriched fractions of CSF derived from sporadic ALS patients

To identify the proteins in the exosome-enriched fractions from CSF that differed between the ALS and iNPH groups, we performed label-free relative quantitation of MS data, as described in the Materials and Methods section. In total, we identified 25,900 spectra and 334 proteins in the exosome-enriched fractions of CSF derived from three ALS and three iNPH patients. Exosome marker proteins including Alix, CD9, and CD81 were accurately identified (Supplementary Table 3). Gene Ontology analysis of the identified proteins in the ALS and iNPH samples using DAVID (v6.8) revealed high enrichment of proteins associated with extracellular vesicles and exosomes (Fig. 1C). The proteins that showed statistically significant differences ($p < 0.05$) between the ALS and iNPH groups are listed in Table 2. Among these, the protein with the greatest increase in the ALS group was nucleolar complex protein 2 homolog, called NIR [novel INHAT (inhibitor of histone acetyltransferases) repressor, $p = 0.0035$, maximum fold change = 2.39].

4.3. Immunoelectron microscopy analysis of CSF exosomes

To confirm that NIR and other candidate proteins specific to ALS were actually contained in the CSF exosomes, we performed immunoelectron microscopy of exosome-enriched samples using EVSecond isolation, followed by ultracentrifugation. Electron microscopy revealed that the exosome-enriched fractions from CSF contained vesicles of 30 to 150 nm in diameter, which were immunopositive for the exosome markers CD81 and CD9 (Fig. 1D and E). We also examined whether these vesicles contained the candidate proteins listed in Table 2 by immunoelectron microscopy. The findings showed that the exosome-like vesicles were positive for anti-NIR antibody as well as exosome marker proteins (Fig. 1F). In contrast, the immunoreactivities of anti-versican core protein antibody and anti-folate receptor alpha antibody were more pronounced in nonexosomal areas than in exosome-like vesicles. We also failed to confirm the immunoreactivities of anti-receptor-type tyrosine-protein phosphatase zeta antibody and anti-charged multivesicular body protein 4a antibody by electron microscopy. We did not investigate the other proteins because no antibodies were available for them, or they were known as nonspecific proteins. Therefore, we focused on the association of NIR and ALS pathology in this study.

4.4. Immunohistochemical analysis of NIR pathology

We immunohistochemically analysed NIR pathology of the motor neurons in the anterior horn of the sporadic ALS patients and disease controls (Supplementary Table 1). In controls, there was typically strong staining for NIR in both the nucleus and cytoplasm (Fig. 2A and B). However, this staining was reduced in the nucleus or in whole cells from ALS patients (Fig. 2C–F). The proportion of nuclei with NIR staining was significantly reduced in ALS patients (Fig. 2G).

ALS patient 6 with upper limb onset showed more severe motor neuron pathology in the cervical and thoracic spinal cord than in the lumbar spinal cord. In parallel with this, we found that NIR staining was more markedly reduced in the motor neurons of the cervical (Fig. 2H and I) and thoracic spinal cord (Fig. 2J and K) than in the lumbar spinal cord (Fig. 2L and M).

We also investigated whether nuclear exclusion of NIR was correlated with cytoplasmic aggregate formation of TDP-43 by double immunofluorescence staining for NIR and TDP-43. We observed that some motor neurons with nuclear exclusion of NIR exhibited TDP-43 aggregates but others did not, and a portion of motor neurons with NIR nuclear retention exhibited TDP-43 aggregates

Table 2

Proteins with statistically significantly increased or decreased normalised abundance in exosome-enriched fractions from CSF of ALS patients are shown by p-value rank (p < 0.05, lowest to highest).

Protein	Gene name	Uniprot accession	ANOVA (p)	Max. fold change	Unique peptides
Increased proteins in ALS samples					
Nucleolar complex protein 2 homolog	<i>NOC2L</i>	Q9Y3T9	0.0035	2.39	1
Programmed cell death 6-interacting protein	<i>PDCD6IP</i>	Q8WUM4	0.015	1.39	1
Versican core protein	<i>VCAN</i>	P13611	0.038	1.36	25
Decreased proteins in ALS samples					
Alpha-1-antichymotrypsin	<i>SERPINA3</i>	P01011	0.0004	3.73	2
Receptor-type tyrosine-protein phosphatase zeta	<i>PTPRZ1</i>	P23471	0.010	2.49	5
Complement C1q subcomponent subunit C	<i>C1QC</i>	P02747	0.017	2.05	2
Coiled-coil domain-containing protein 19, mitochondrial	<i>CCDC19</i>	Q9UL16	0.025	3.02	1
Myosin light chain 6B	<i>MYL6B</i>	P14649	0.027	2.37	1
Macrophage receptor MARCO	<i>MARCO</i>	Q9UEW3	0.027	1.70	1
IgG Fc-binding protein	<i>FCGBP</i>	Q9Y6R7	0.028	1.77	82
Folate receptor alpha	<i>FOLR1</i>	P15328	0.033	2.08	1
Reelin	<i>RELN</i>	P78509	0.038	2.45	8
Complement factor B	<i>CFB</i>	P00751	0.039	2.27	1
Charged multivesicular body protein 4a	<i>CHMP4A</i>	Q9BY43	0.042	1.84	1

(Supplementary Fig. 2), indicating that nuclear exclusion of NIR was irrelevant to TDP-43 aggregate formation.

5. Discussion

In this study, we obtained a comprehensive overview of the proteins in exosome-enriched fractions of CSF derived from ALS patients by proteomic analysis and focused on NIR, the protein showing the greatest increase in exosome-enriched fractions from CSF of the ALS group compared with the level in controls. Interestingly, the subcellular distribution of NIR was altered in ALS motor neurons, suggesting that NIR may play a role in the pathogenesis of ALS.

Most previous proteomic studies of CSF exosomes (Chiasserini et al., 2014; Lee et al., 2016; Manek et al., 2018; Street et al., 2012; Welton et al., 2017) used pooled CSF or continuously drained CSF because of the extremely small amounts of CSF exosomes contained in the volume of CSF normally available from an individual. In this study, we confirmed the feasibility of several methods for isolating CSF exosomes efficiently from a small volume (3.5 ml) of CSF for high-quality analysis in LC/MS-MS. The exosome-enriched fractions isolated using the EVSecond device were confirmed to be positive for the exosome marker proteins CD9 and CD81 by western blot analysis and by immunoelectron microscopy (Fig. 1). All proteins identified by proteomic analysis (Table 2) were already listed in Vesiclepedia, the extracellular vesicle database (www.microvesicles.org/), with the exception of the macrophage receptor MARCO, indicating the validity of our method for identifying exosome proteins. Interestingly, those proteins were not included among the results from a previous CSF (not CSF exosomes) proteomic study of ALS patients (Chen et al., 2016; Collins et al., 2015). Therefore, our proteomic data appear to be unique to CSF exosomes and are potentially valuable in the search for ALS biomarkers. Unexpectedly, we failed to detect TDP-43, probably because of its low peptide concentration below the cut-off used in our analysis.

Proteomic analysis revealed that the level of NIR was significantly increased in exosome-enriched fractions from CSF of the ALS group; its localisation in exosome-like vesicles of CSF was also confirmed by immunoelectron microscopy. Because NIR was abundantly expressed in motor neurons of disease controls (Fig. 2A and B), an increased level of exosomal NIR at a relatively early disease stage (ALS-Functional Rating Scale 41–43, Table 1) could be derived from motor neurons. However, the contribution of cell types other than motor neurons could not be completely excluded. We also

found that NIR was reduced in the nucleus of motor neurons in end-stage sporadic ALS patients. Although some reports have indicated the association of NIR with cancer (Klein et al., 2018; Titus et al., 2017), the present study provides the first evidence of NIR involvement in human neurological disease. In normal conditions, NIR is reported to be located predominantly in the nucleolus of several cultured cell lines, and it translocates from the nucleolus to the nucleoplasm in response to the nucleolar stress imposed by, for example, actinomycin D (Heyne et al., 2014). Nucleolar stress is characterised by the mislocalisation of nucleolar proteins and abnormalities in rRNA transcription and processing. In recent years, there has been increased interest in the role of nucleolar stress in the pathogenesis of neurodegenerative diseases (Herrmann and Parlato, 2018; Mizielinska et al., 2017; Tao et al., 2015). For example, nucleolar dysfunction was observed in C9orf72-associated ALS or frontotemporal lobar degeneration, although the changes and functions of nucleolar proteins in sporadic ALS remain unclear. Several nucleolar proteins play important roles in the MDM2-p53 regulatory pathway in response to nucleolar stress (Jin et al., 2004; Lohrum et al., 2003; Xie et al., 2016; Zhang et al., 2003). As one of the nucleolar proteins, NIR was reported to bind to p53 and inhibit p53-mediated gene transactivation by blocking histone acetylation and to support the MDM2-mediated repression of p53 by binding to MDM2, the E3 ubiquitin ligase (Heyne et al., 2014). Accordingly, knockdown of NIR increases p53-dependent apoptosis (Hublitz et al., 2005). Given our knowledge about the pathogenesis of ALS, it is hypothesised from our observations that the reduction of NIR in motor neurons, possibly triggered by nucleolar stress-induced NIR translocation from the nucleolus and release into CSF presumably via exosomes, may affect apoptotic neuronal death in ALS. We suggest that NIR could serve as a good biomarker for the perturbation of nucleolar homeostasis in ALS, although further research is needed to explore a direct link between elevated NIR in CSF exosomes and decreased expression in motor neurons. Furthermore, elucidation of the role of NIR in motor neurons opens a new avenue for the understanding of the pathological mechanisms of ALS.

This study had some limitations. The proteomic control data might have been affected by iNPH-related conditions because of the use of disease controls, and we failed to quantify NIR in CSF exosomes by biochemical methods including western blotting because of their low concentrations. We continue our efforts to measure the very small amount of protein from CSF exosomes.

In conclusion, we have provided the first protein profiling data of exosome-enriched fractions from CSF derived from ALS patients

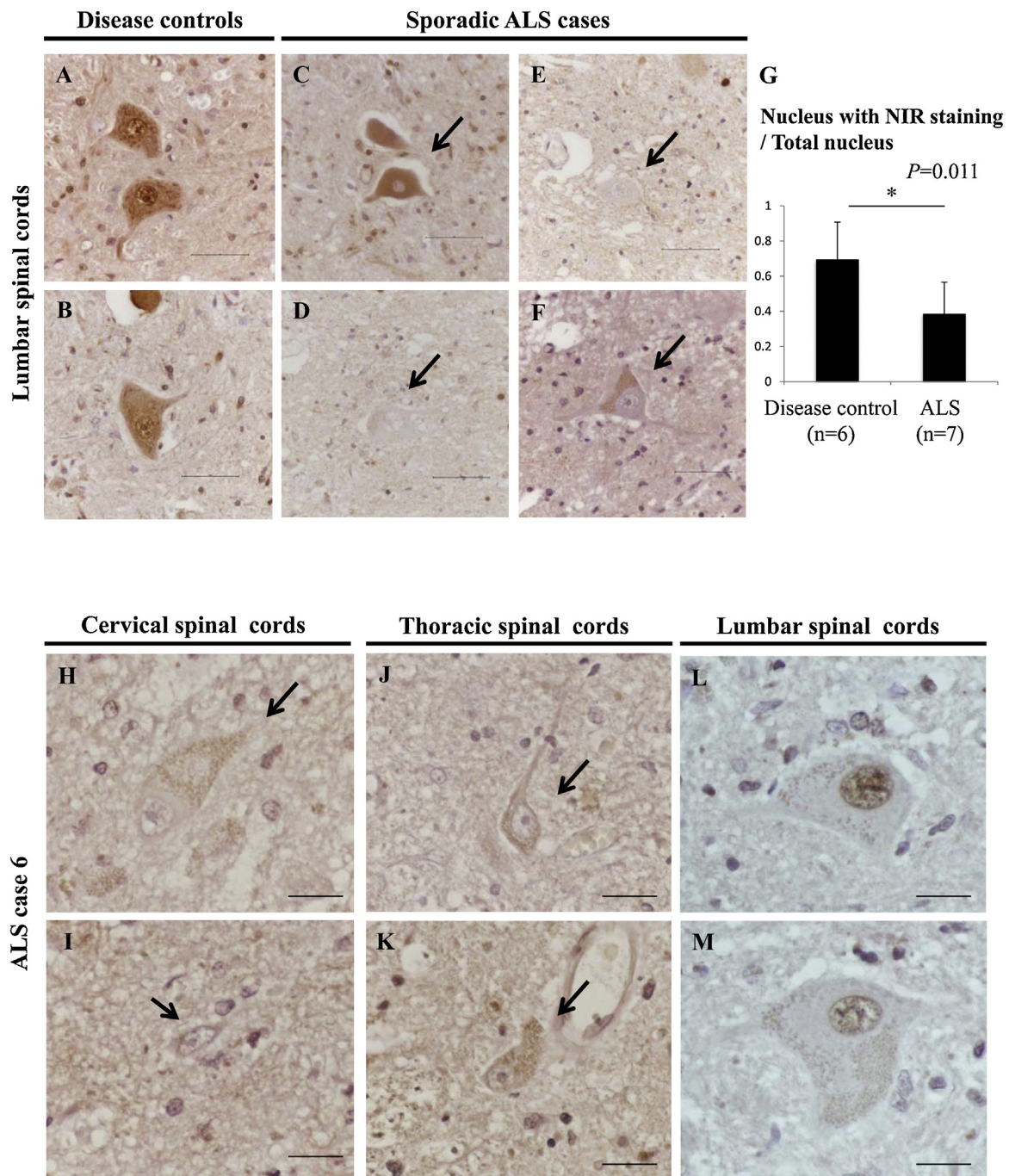


Fig. 2. Immunohistochemical analysis of NIR pathology.

(A–G) The images show the NIR staining patterns of the lumbar spinal cord in sporadic ALS and disease control patients. A and B: Immunohistochemical staining with anti-NIR antibody in the motor neurons of control case 8 (A) and case 9 (B). C–F: Immunohistochemical staining with anti-NIR antibody in the motor neurons of ALS case 1 (C), case 2 (D and E), and case 7 (F). Arrows indicate the motor neurons with reduced NIR staining in the nucleus (C) or in whole cells (D–F). Scale bar: 50 μ m. The frequency of nuclei with NIR staining among the total nuclei was determined in two sections of the thoracic or lumbar spinal cord (G). $*p < 0.05$.

(H–M) Differential staining patterns of NIR based on the severity of motor neuron degeneration in sporadic ALS case 6. The motor neurons of the cervical spinal cord (H and I) and thoracic spinal cord (J and K) have severe motor neuron pathology, whereas the motor neurons of the lumbar spinal cord (L and M) have mild pathology. Scale bar: 20 μ m.

by proteomic analysis. Our established methodology to isolate exosomes from small amounts of CSF and to purify peptides for high-quality proteomic analysis should be applicable to the search for disease-related proteins in larger-scale samples or in various neurological diseases. We identified NIR as the protein with the greatest increase in exosome-enriched fractions from CSF of ALS patients, consistent with the pathologically confirmed depletion of NIR from motor neurons, although its role in ALS pathogene-

sis remains to be determined. Further investigations are required to elucidate the pathophysiological significance of the association between NIR and ALS.

Declarations of interest

None.

Acknowledgments

This work was supported in part by a grant of Creation and Innovation Centers for Advanced Interdisciplinary Research Areas (H.H.); Grants-in-Aid for Scientific Research (H.D.: 18K07503; K.Tan.: 18K07504, and F.T.: 18K07532) from the Japan Society for the Promotion of Science; the Takeda Science Foundation (H.D.); and a Grant for Strategic Research Promotion from Yokohama City University (F.T.:# SK2804). The MS analysis was performed at Clinical Mass Spectrometer Platform (Yokohama City University) supported by the Ministry of Education, Culture, Sports, Science and Technology, Japan. We thank Edanz (www.edanzediting.co.jp) for editing the English text of a draft of this manuscript.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.neures.2019.10.010>.

References

- Brooks, B.R., Miller, R.G., Swash, M., et al., 2000. El Escorial revisited: revised criteria for the diagnosis of amyotrophic lateral sclerosis. *Amyotroph. Lateral Scler. Other Motor Neuron Disord.* 1, 293–299.
- Chen, Y., Liu, X.H., Wu, J.J., et al., 2016. Proteomic analysis of cerebrospinal fluid in amyotrophic lateral sclerosis. *Exp. Ther. Med.* 11, 2095–2106.
- Chiasserini, D., van Weering, J.R., Piersma, S.R., et al., 2014. Proteomic analysis of cerebrospinal fluid extracellular vesicles: a comprehensive dataset. *J. Proteomics* 106, 191–204.
- Collins, M.A., An, J., Hood, B.L., et al., 2015. Label-free LC-MS/MS proteomic analysis of cerebrospinal fluid identifies Protein/Pathway alterations and candidate biomarkers for amyotrophic lateral sclerosis. *J. Proteome Res.* 14, 4486–4501.
- Ding, X., Ma, M., Teng, J., et al., 2015. Exposure to ALS-FTD-CSF generates TDP-43 aggregates in glioblastoma cells through exosomes and TNTs-like structure. *Oncotarget* 6, 24178–24191.
- Grey, M., Dunning, C.J., Gaspar, R., et al., 2015. Acceleration of alpha-synuclein aggregation by exosomes. *J. Biol. Chem.* 290, 2969–2982.
- Herrmann, D., Parlato, R., 2018. C9orf72-associated neurodegeneration in ALS-FTD: breaking new ground in ribosomal RNA and nucleolar dysfunction. *Cell Tissue Res.*
- Heyne, K., Forster, J., Schule, R., et al., 2014. Transcriptional repressor NIR interacts with the p53-inhibiting ubiquitin ligase MDM2. *Nucleic Acids Res.* 42, 3565–3579.
- Hublitz, P., Kunowska, N., Mayer, U.P., et al., 2005. NIR is a novel INHAT repressor that modulates the transcriptional activity of p53. *Genes Dev.* 19, 2912–2924.
- Iguchi, Y., Eid, L., Parent, M., et al., 2016. Exosome secretion is a key pathway for clearance of pathological TDP-43. *Brain* 139, 3187–3201.
- Jin, A., Itahana, K., O'Keefe, K., et al., 2004. Inhibition of HDM2 and activation of p53 by ribosomal protein L23. *Mol. Cell. Biol.* 24, 7669–7680.
- Kawakami, K., Fujita, Y., Matsuda, Y., et al., 2017. Gamma-glutamyltransferase activity in exosomes as a potential marker for prostate cancer. *BMC Cancer* 17, 316.
- Klein, A.P., Wolpin, B.M., Risch, H.A., et al., 2018. Genome-wide meta-analysis identifies five new susceptibility loci for pancreatic cancer. *Nat. Commun.* 9, 556.
- Lee, J., McKinney, K.Q., Pavlopoulos, A.J., et al., 2016. Exosomal proteome analysis of cerebrospinal fluid detects biosignatures of neuromyelitis optica and multiple sclerosis. *Clin. Chim. Acta* 462, 118–126.
- Lohrum, M.A.E., Ludwig, R.L., Kubbutat, M.H.G., et al., 2003. Regulation of HDM2 activity by the ribosomal protein L11. *Cancer Cell* 3, 577–587.
- Manek, R., Moghieb, A., Yang, Z., et al., 2018. Protein biomarkers and neuroproteomics characterization of microvesicles/exosomes from human cerebrospinal fluid following traumatic brain injury. *Mol. Neurobiol.* 55, 6112–6128.
- Mizielinska, S., Ridler, C.E., Balendra, R., et al., 2017. Bidirectional nucleolar dysfunction in C9orf72 frontotemporal lobar degeneration. *Acta Neuropathol. Commun.* 5, 29.
- Patel, G.K., Khan, M.A., Zubair, H., et al., 2019. Comparative analysis of exosome isolation methods using culture supernatant for optimum yield, purity and downstream applications. *Sci. Rep.* 9, 5335.
- Saman, S., Kim, W., Raya, M., et al., 2012. Exosome-associated tau is secreted in tauopathy models and is selectively phosphorylated in cerebrospinal fluid in early Alzheimer disease. *J. Biol. Chem.* 287, 3842–3849.
- Street, J.M., Barran, P.E., Mackay, C.L., et al., 2012. Identification and proteomic profiling of exosomes in human cerebrospinal fluid. *J. Transl. Med.* 10, 5.
- Stuendl, A., Kunadt, M., Kruse, N., et al., 2016. Induction of alpha-synuclein aggregate formation by CSF exosomes from patients with Parkinson's disease and dementia with Lewy bodies. *Brain* 139, 481–494.
- Tao, Z., Wang, H., Xia, Q., et al., 2015. Nucleolar stress and impaired stress granule formation contribute to C9orf72 RAN translation-induced cytotoxicity. *Hum. Mol. Genet.* 24, 2426–2441.
- Taylor, J.P., Brown Jr., R.H., Cleveland, D.W., 2016. Decoding ALS: from genes to mechanism. *Nature* 539, 197–206.
- Thery, C., Amigorena, S., Raposo, G., et al., 2006. Isolation and characterization of exosomes from cell culture supernatants and biological fluids. *Curr. Protoc. Cell Biol.*, Chapter 3, Unit 3.22.
- Titus, A.J., Way, G.P., Johnson, K.C., et al., 2017. Deconvolution of DNA methylation identifies differentially methylated gene regions on 1p36 across breast cancer subtypes. *Sci. Rep.* 7, 11594.
- Welton, J.L., Loveless, S., Stone, T., et al., 2017. Cerebrospinal fluid extracellular vesicle enrichment for protein biomarker discovery in neurological disease; multiple sclerosis. *J. Extracell. Vesicles* 6, 1369805.
- Xie, N., Ma, L., Zhu, F., et al., 2016. Regulation of the MDM2-p53 pathway by the nucleolar protein CSIG in response to nucleolar stress. *Sci. Rep.* 6, 36171.
- Yuyama, K., Sun, H., Mitsutake, S., et al., 2012. Sphingolipid-modulated exosome secretion promotes clearance of amyloid-beta by microglia. *J. Biol. Chem.* 287, 10977–10989.
- Yuyama, K., Sun, H., Usuki, S., et al., 2015. A potential function for neuronal exosomes: sequestering intracerebral amyloid-beta peptide. *FEBS Lett.* 589, 84–88.
- Zhang, Y., Wolf, G.W., Bhat, K., et al., 2003. Ribosomal protein L11 negatively regulates oncoprotein MDM2 and mediates a p53-Dependent ribosomal-stress checkpoint pathway. *Mol. Cell. Biol.* 23, 8902–8912.